

Renchi Zhang

renchi.rz.zhang@gmail.com | +31 (0)645404998 | Delft, the Netherlands | [GitHub](#) | [LinkedIn](#) | [Google Scholar](#)

PhD researcher in Cognitive Robotics (TU Delft, defending Sept. 2026) who builds robot systems driven by foundation models and measures where they work and where they fail. Has taken an LLM planner onto a Boston Dynamics Spot and evaluated it against rule-based constraints, integrated a manipulation stack on a Franka Emika Panda with perception pre-trained on Isaac synthetic data and fine-tuned on real captures, and run in-lab robot experiments with 218 operators over 872 trials. Evaluation methodology is a through-line: controlled benchmarks with human baselines (6,300 stimuli, 1,250 participants), failure-mode taxonomies, and LLM-assisted annotation validated against human coding. Currently also fine-tuning a vision-language-action policy (NVIDIA Isaac-GR00T) to test how robust it is to variations in language commands and in the visual properties of the scene. Four first-author studies, three in peer-reviewed journals; every system open-sourced. Based in Delft, Dutch permanent resident; available from Oct. 2026.

EDUCATION

Delft University of Technology (TU Delft) | Ph.D. in Cognitive Robotics 2022.07–2026.09 (expected)
Thesis: information-processing alignment between humans and LLM/VLM-driven robots across perception, reasoning and response.

Leiden University | M.Sc. in Computer Science (Artificial Intelligence, cum laude) 2019.09–2021.11
Thesis: modelling bi-directional hysteresis with an LSTM to navigate a robotic catheter for cardiac catheterization.

Katholieke Universiteit Leuven | Visiting student, surgical robotics 2021.09–2021.10

Beijing University of Posts and Telecommunications (BUPT) | B.Eng. in Digital Media Technology 2014.09–2018.06

SELECTED PROJECTS

LLM Planner for Robot Navigation, Executed on Boston Dynamics Spot | [Pre-print \[5\]](#) | [Demo Video](#) | [Code](#) 2024–2026

- **Model-to-robot execution path:** a system prompt carrying role, a structured environment description (grid geometry, obstacles, checkpoints), hard constraints, a strict robot-executable output schema and a refusal guardrail for infeasible requests; grid-to-world-frame transformation and execution on Spot via ROS and the Spot SDK, plus a natural-language grasp driven by a multimodal model.
- **Evaluation protocol:** rule-based scoring with per-violation penalties (collisions, illegal moves, out-of-bounds, revisits); penalty-free and no-plan rates as headline metrics. Reasoning vs. non-reasoning planner on identical tasks: 55.7% vs. 18.7% penalty-free, at 1.2 s / 1,032 tokens rising to 51.6 s / 7,167 tokens; 19.8% of the hardest requests returned no plan. Failure modes classified by type and horizon.
- **Platform:** host/client web service with parallel inference, automatic scoring and trajectory visualisation; 226 users, 10 sessions, 494 requests. Prompt strategies coded with an 11-category codebook and correlated with scores.

Controlled Benchmark with Human Baselines for MLLM Perception | [Paper \[1\]](#) | [Demo Site](#) | [Code](#) 2024–2026

- **Benchmark design:** a 6,300-image benchmark with a reproducible generation pipeline: 2D scenes plus Blender/CLEVR 3D renders with occlusion, shading and perspective; orthogonal manipulation of distractor heterogeneity and scene structure; 9 conditions; set sizes 4 to 256.
- **Evaluation protocol:** 1,250 human participants (157,042 trials) as the baseline; three MLLMs queried 10 times per image at temperature 0 to control response stochasticity; Spearman correlations across conditions.
- **Finding:** difficulty rankings agree strongly ($\rho = 0.82$ on error rate, $\rho = 0.91$ against human response time), but the models lack a quitting threshold and collapse to a single answer under high distractor heterogeneity and 3D occlusion. Aggregate accuracy masks the mechanism difference.

OpenChessRobot: Reproducible Chess-Playing Manipulator for HRI Research | [Paper \[3\]](#) | [Demo Video](#) | [Code](#) 2022–2025

- **Sim-to-real perception:** ResNet-34 / InceptionV3 occupancy and piece classifiers pre-trained on NVIDIA Isaac synthetic scenes (5,000 positions with varied lighting, camera poses and backgrounds) and fine-tuned on robot-collected real captures; 96.5–100% piece recognition. A state-legality check triggers re-imaging from a shifted camera pose instead of acting on an implausible state.
- **Manipulation stack:** libfranka and franka_ros on a real-time kernel, MoveIt! motion planning, ArUco 3D localisation refined by non-linear least squares, a custom compliant gripper. A full 97-move game was instrumented (~7.3 s per move); a grasp study across piece sizes and offsets localised residual error to the coordinate transform rather than the gripper.
- **Interaction and user study:** LLM commentary grounded in the perceived board state; 597-participant online study across three scenarios; expert (FIDE Master) case study.

Voice vs. Gesture Control of a Quadruped: Operator Study | [Paper \[2\]](#) | [Demo Video](#) | [Code](#) 2023–2025

- **Interface:** HoloLens 2 application (Unity, MRTK) streaming hand-skeleton and 6-DoF headset pose data at ~30 Hz to a ROS backend; SVM gesture classifier stabilised by sliding-window voting, confidence gating and a post-command lockout; robot odometry and headset pose logged for every trial.

- **Experiment:** 2×2 within-subjects design, 218 operators, 872 trials. An exocentric viewpoint raised left/right reversal errors $6.7\times$ under gesture control and not at all under speech; operator-robot bearing alignment was quantified from odometry and headset pose.
- **Annotation validation:** 217 exit interviews transcribed with Whisper and coded by a reasoning LLM, checked against manual counts ($r = 0.98$ over 8 categories, one off by $\sim 40\%$).

VLA Fine-Tuning and Robustness to Command and Scene Variation (NVIDIA Isaac-GR00T)

2026.02–Present

- **Research question:** how robust a fine-tuned vision-language-action policy is to rephrased language commands and to changes in the visual properties of the scene, i.e. whether it learned the task or the surface form of its training data.
- **Training pipeline:** fine-tuned Isaac-GR00T (behaviour cloning from demonstration data) with LoRA on multi-GPU DDP, single-node in the lab and multi-node on TU Delft's SLURM cluster; built the data-formatting and experiment-tracking pipeline (DVC, WandB), debugged NCCL, optimizer-state memory and IO bottlenecks, and worked with FlashAttention and DeepSpeed for memory-limited runs. Evaluation so far is offline.

Evaluation Datasets for a Human-AI Hybrid System | MARS V Fellow, Cambridge AI Safety Hub

2026.06–Present

- **Dataset construction:** building benchmarks for an LLM-judge router that assigns tasks to humans or models (project initiated by two former Google DeepMind researchers, Sampura Research). Tasks must satisfy three criteria: model-alone accuracy below 70%, reliable ground truth, and human solvability. Redesigned a saturated chain-of-thought monitoring set (0.94–0.96) so that the gate model gpt-5.4-mini scores 0.645 on a balanced set, leaving headroom for validation.

PUBLICATIONS

- [1] **Zhang, R.**, de Winter, J. C. F., Dodou, D., Seyffert, H., & Eisma, Y. B. (2026). Human-like attention? A psychophysical comparison of visual search in humans and MLLMs. *Computational Brain & Behavior*. [DOI](#) · [Code](#)
- [2] **Zhang, R.***, van der Linden, J.*, Dodou, D., Seyffert, H., Eisma, Y. B., & de Winter, J. C. F. (2025). Walk along: An experiment on controlling the mobile robot 'Spot' with voice and gestures. *ACM Transactions on Human-Robot Interaction*, 14(4), Article 64. [DOI](#) · [Code](#) · * Equal contribution
- [3] **Zhang, R.**, de Winter, J. C. F., Dodou, D., Seyffert, H., & Eisma, Y. B. (2025). An open-source reproducible chess robot for human-robot interaction research. *Frontiers in Robotics and AI*, 12, Article 1436674. [DOI](#) · [Code](#)
- [4] Wu, D.*, **Zhang, R.***, Pore, A., Dall'Alba, D., Ha, X. T., Li, Z., ... & Vander Poorten, E. (2024). A review on machine learning in flexible surgical and interventional robots: Where we are and where we are going. *Biomedical Signal Processing and Control*, 93, Article 106179. [DOI](#) · * Equal contribution; compiled the review's reinforcement-learning methods
- [5] **Zhang, R.**, de Winter, J. C. F., Dicke, T., Dodou, D., & Eisma, Y. B. (2026). Educating future engineers about LLMs: A scalable workshop. Under review. [Pre-print](#) · [Code](#)

EXPERIENCE

MARS V Fellow Cambridge AI Safety Hub, UK / remote, part-time	2026.06–Present
PhD Researcher (employed) Cognitive Robotics, TU Delft	2022.07–2026.09
Workshop Instructor TU Delft; 10 sessions of an LLM-robotics workshop, 226 MSc students	2025.12–2026.01
MSc Thesis Co-supervisor TU Delft; pose estimation, MLLM manipulation, human-robot handover	2023–2026
Teaching Assistant TU Delft (Human-Robot Interaction) and Leiden University (Deep Learning)	2023–2024, 2020–2021
Product Manager Intern BMW China, Beijing; IT department	2017.06–2017.11

TECHNICAL SKILLS

Machine learning and models: PyTorch, HuggingFace, Transformer / LLM / VLM fundamentals; LLM planners with schema-constrained output and guardrails; CNN perception (ResNet, Inception); LSTM sequence modelling; LoRA fine-tuning of a VLA policy (NVIDIA Isaac-GR00T) with multi-GPU and multi-node DDP (torchrun, SLURM), FlashAttention, DeepSpeed (working knowledge), DVC, WandB, NCCL / optimizer-state / IO debugging. Reinforcement learning at literature level (compiled the RL methods of review [4]) plus coursework.

Robotics and simulation: ROS Noetic, Franka Emika Panda (libfranka, franka_ros), Boston Dynamics Spot SDK, MoveIt! (IKFast, RRTConnect), NVIDIA Isaac Sim, Gazebo, Blender/CLEVR synthetic data, real-time kernel setup, ArUco and OpenCV calibration and 3D localisation, stereo camera (ZED2), markerless hand and head pose (HoloLens 2), eye-tracking (Tobii Pro Glasses 3).

Data and evaluation: benchmark and stimulus-set design, human-baseline collection (Prolific, Qualtrics), within- and between-subjects experiment design, failure-mode taxonomies, rule-based scoring, LLM-assisted annotation validated against human coding, Spearman correlation and rank-transform robustness checks, binomial confidence intervals.

Engineering: Python (primary), C++ (ROS nodes), C# (Unity), Linux, shell, Git, CUDA; concurrent inference serving; English technical writing (five papers).